CO-2

ASSESSMENT-3

CLASS TEST

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CO2_AT_3 Class Test

1. A dark museum photograph can be enhanced using logarithmic or power-law (gamma) transformation

Processing steps:

1. Read the image and convert it to greyscale.
2. Apply gamma correction using $S = Cr^{\gamma}$, with $\gamma < 1$, to brighten dark regions.
3. Alternatively, apply logarithmic transformation $S = c\log(1+r)$ to expand dark intensity values.
4. Normalize the output intensity values and display the enhanced images.

Justification :- Gamma correction brightens shadow regions, while log transformation improves visibility of dark details. Both help reveal hidden exhibit details.

Results- The enhanced image shows improved brightness and visibility in shadow regions.

2. Histogram equalization improves the contrast of finger print ridges and valleys when pixels intensities occupy a narrow range.

Processing steps:

1. Convert the fingerprint image to grayscale.
2. Calculate the histogram of pixel intensities
3. Compute the normalized histogram: $p(r_k) = \frac{n_k}{MN}$
4. Calculate the cumulative distribution function (CDF):
5. Map each intensity using: $s_k = \text{round}\left[(L-1)\, C(r_k)\right]$.
6. Display the enhanced fingerprint image.

Justification: The intensity mapping redistributes pixel values across a wider range, improving contrast between ridges and valleys.

Result: Fingerprint ridges become clearer and easier to identify.

3. Salt and Pepper noise ~~red~~ produces random black and white pixels. A median filter removes these outliers while preserving edges.

Processing steps:

1. Read the noisy image.
2. Select a 3×3 neighborhood around each pixel.
3. Sort the pixel values and calculate their median
4. Replace the center pixel with median value.
5. Repeat for all pixels.

Justification:- Median filtering removes extreme values without averaging all neighboring pixels. An averaging filter may blur object boundaries and spread noise.

Result:- The image becomes cleaner while retaining important edges.

4. Laplacian sharpening enhances copper-track edges in slightly blurred circuit board images.

Processing steps:

1. Read the image and convert it to grayscale.
2. Apply mild Guassian smoothing if noise is present.
3. Calculate the laplacian using a second order derivative filter.
4. Sharpen the image using:

$$g(x,y) = f(x,y) - \nabla^2 f(x,y)$$

This formula assumes a standard Laplacian sign convention.

5. Display and compare the sharpened image.

Justification :- The Laplacian highlights rapid intensity changes around edges, improving the visibility of copper tracks.

Result :- Track boundaries and fine details becomes sharper.

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5. Frequency domain low-pass filtering reduces grainy high-frequency noise in drone images.

Processing Steps:

1. Convert the images to grayscale.
2. Apply the Fourier Transform (FFT)
3. Design a low-pass filter to retain low frequencies and suppress high frequencies.
4. Multiply the frequency spectrum by the filter.
5. Apply the inverse Fourier Transform to obtain the filtered image.

Justification :- A Guassian or Butterworth low-pass filter can reduce noise while controlling the amount of smoothing.

Result :- The image becomes smoother, making farm-plot outlines easier to inspect.

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6. High pass or high boost filtering enhances fine details and vessel edges in retinal fundus images.

Processing steps :-

1. Convert the retinal image to grayscale.

2. Apply the Fourier Transform.
3. Design a high-pass filter to suppress low frequencies and retain high frequencies.
4. Multiply the spectrum by the filter and apply the inverse Fourier Transform.
5. Alternatively, use high-boost filtering to enhance details while preserving the original image.

Justification: High-pass filtering emphasizes rapid intensity changes, while high-boost filtering enhances details without completely removing original image information.

Result:- Thin vessel edges and fine retinal details become more visible, although excessive filtering may amplify noise.

7. Point and line detection identify air bubbles and hairline cracks on glass bottles.

Processing steps:-

1. Convert the bottle image to grayscale.
2. Apply mid noise reduction.
3. Use a point detection mask:

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

4. Apply directional line masks to detect horizontal, vertical and diagonal cracks.
5. Threshold the filter response and mark detected defects.

Result:- Isolated bubbles and thin cracks can be detected and outlined.

8. Thresholding seperates dark handwriting from the white paper background.

Processing steps:

1. Convert the answer-sheet image to grayscale.
2. Apply mild filtering to reduce noise.
3. Select a thresholding method.
4. For uneven lighting, apply adaptive thresholding, which calculates a local threshold for each neighborhood.
5. Convert the image into a binary image with black handwriting and a white background.

Justification:- Global thresholding uses one threshold for the entire image and works well under uniform illumination. Adaptive thresholding handles local brightness variations and is more suitable for unevenly lit answer sheets.

Result:- Handwriting is seperated from the background for further processing.

9. Lake extraction Using Region Growing

Region growing extracts a lake by grouping connected pixels with similar intensity or color.

Processing steps:-

1. Read the satellite images and apply mild noise reduction.

2. Select a seed pixel inside the lake.
3. Define a similarity criterion:

$$|I(x,y) - \mu_R| < T$$

4. Add neighbouring pixels that satisfy the criterion.
5. Continue until no neighboring pixels meets the condition.
6. Apply morphological operations to smooth the extracted boundary

Result:- The lake is extracted as a connected region from the satellite image.

10. Edge linking connects broken edge segments to obtain a complete gear outline.

Processing steps:-

1. Convert the gear image to grayscale.
2. Apply Gaussian smoothing to reduce noise.
3. Use the canny edge detector to identify edges.
4. Apply local edge linking or morphological closing to connect small gaps.
5. Use the Hough transform to detect suitable geometric structures.
6. Apply contour tracing to extract the complete gear boundary.

Justification: Local edges linking connects nearby edges with similar directions, while morphological closing bridges small gaps. Hough transform helps detect geometric structures.

Result:- A more complete gear outline is obtained for inspection.